CoachPulse

Features and UX/UI overview

A business operating system for independent personal trainers: client CRM, workout programming, scheduling and a real double-entry accounting ledger — offline-first, no card gateway.

Phase 1 complete · 8 of 8 milestones · Go API + Expo app

1. Product summary

CoachPulse serves independent personal trainers who settle in cash, local bank wires and P2P transfers — so there is deliberately **no card gateway**. Trainers get one app for their client roster, programming, calendar and books, and it keeps working in basements and gyms with no signal.

Its distinguishing idea is that the accounting is **real**: prepaid packs are a liability (Deferred Revenue) and revenue is recognised only when a session is delivered. The P&L, receivables ageing and tax export all derive from one immutable journal, so they agree by construction.

Fact	Value
Backend	Go 1.24, chi, sqlc, Postgres 16, S3-compatible media storage (MinIO)
Client	One Expo / React Native codebase for iOS, Android and web; local SQLite mirror
Isolation	Postgres row-level security on all 35 tables; API role is neither superuser nor BYPASSRLS
Scale of build	~12,700 lines production Go, 186 server tests, 74 offline client tests, 14 client routes

2. Feature set

Foundation & security

- Sign-up creates a trainer and business, seeding a chart of accounts and a library of **66 exercises** so nothing starts empty.
- JWT access tokens with rotating refresh tokens tracked as a family — replaying a spent token revokes the chain.
- Structured errors with stable machine codes; logger redacts medical notes and payment details; UUIDv7 ids; integer minor-unit money that refuses cross-currency arithmetic.
- Idempotency-Key on every mutating endpoint: replaying an outbox batch produces exactly one journal entry.

Double-entry ledger

- Accounts, journal entries and lines; entries are append-only, corrections are reversing entries.
- A deferred constraint trigger rejects any unbalanced entry at commit.
- Single posting funnel (`ledger.Post`); reports: P&L, trial balance, receivables ageing, tax-ready export.

Event	Debit	Credit
Invoice issued (10-session pack)	Accounts Receivable	Deferred Revenue
Payment recorded	Cash / Bank / Wallet	Accounts Receivable
Session completed	Deferred Revenue	**Training Revenue**
Expense logged	Expense account	Cash / Bank

Client CRM & intake

- Client records with custom tags, searchable roster, archiving.
- PAR-Q health questionnaire, versioned waivers, biometrics history (weight, body fat).
- Medical notes encrypted with pgcrypto and excluded from ordinary profile fetches.

- Progress photos via presigned URLs — bytes never pass through the API; authorisation happens before the URL exists.
- Client portal isolation: policies narrow to a client's own row when a portal id is bound.

Scheduling & attendance

- Session types (name, duration, capacity, credit cost); recurring sessions; calendar over a window.
- Attendance state machine per **attendee**: scheduled, completed, no-show, late cancel, early cancel — so in a semi-private slot each client has their own credit and revenue outcome.
- Bulk roster check-off (mark-day).
- Double-booking refused by a Postgres exclusion constraint over a buffered time range (e.g. 15 min).
- Completing a session burns a credit and recognises revenue; at zero credits the app offers a renewal or an explicit overdraft.

Billing, invoices & payments

- Prepaid packages grant credits and issue an invoice in one step; manual credit grants supported.
- **Gap-free invoice numbers** (row-locked counter, not a sequence), as required by tax law across much of the EU.
- Overdue status is derived, never stored. Payments recorded manually against Cash, Bank or Wallet; overpayment refused.
- Shareable invoice link (32 random bytes, hash stored, revocable) — the only unauthenticated surface — plus branded PDF rendered from the same HTML template via headless Chromium.
- Payment methods per trainer (e.g. IBAN shown on the shared invoice); low-balance detection for renewals.

Programming & workout logging

- 66 exercises per tenant organised by **movement pattern**; mesocycles → blocks → days → prescriptions (a 4-week block repeats days rather than storing 28 copies).
- Prescription and performance are separate facts, preserving progression history.
- Set logging upserts on (workout, exercise, set index) so corrections and outbox replays converge.
- **Repeat last time** clones last session's working sets as logged-but-not-completed; warm-ups skipped; progression counts working sets only.
- Per-lift volume progression history.

Practice dashboard

- The five numbers a trainer opens the app for: earned this month, sessions today, sessions left this week, unpaid invoices, and clients who need renewing.

Offline-first sync

- Pull mirrors rows into local SQLite with per-collection cursors behind an opaque token; every read hits the device.
- Push carries **operations, not rows**, dispatched to the same services as the online path — no route around ledger rules.

- Conflict policy split by data class: last-write-wins for workout logs, biometrics and notes; strict serial intents for money and attendance, surfacing typed conflicts to the trainer.
- Offline-minted UUIDv7 ids are honoured by the server so a device can queue a workout and its sets in one batch.
- Demo mode (`EXPO_PUBLIC_DEMO=1`) seeds a practice locally with no server.

3. Key user journeys

Journey A — a session becomes revenue

- Create account → add client (works instantly, with or without signal).
- Sell a pack (e.g. 10 × 50.00): invoice issued, 10 credits granted. Books: DR Receivable / CR Deferred Revenue — nothing earned yet.
- Book a session (API today; no in-app screen yet).
- Mark Completed on Today: credit burns, balance drops to 9, DR Deferred / CR Training Revenue — revenue recognised at delivery.
- At zero credits the app offers renewal or explicit overdraft; it never trains someone for free silently.

Journey B — getting paid

- Money tab shows the invoice outstanding with ageing.
- Mark as paid → amount prefilled → choose Bank/Cash/Wallet → record.
- Receivables go to zero (DR Bank / CR Receivable); revenue unchanged.

Offline test

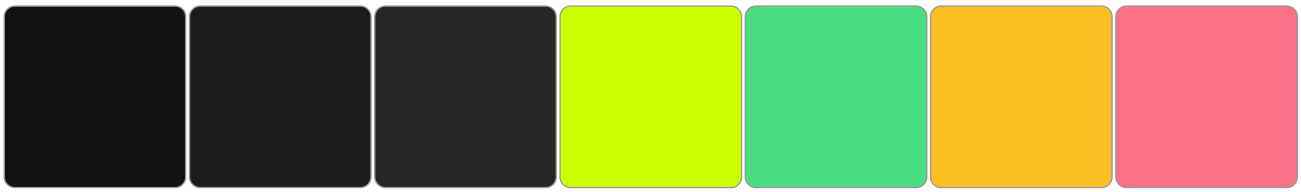
Wifi off: mark attendance, log a workout, record a payment. Everything responds instantly and a badge counts queued work. On reconnect the queue drains; server-owned numbers (credit balances, invoice status) update when the server answers.

4. UX/UI design

Design principles

- **One lime thing per screen** — the accent marks what the trainer came to do; everything else is greyscale, readable in bad gym light.
- **Optimistic about choices, never about what the server owns.** An attendance mark shows instantly; a credit balance or 'settled' status waits for the server.
- **Refusals are shown, not hidden.** A rejected operation stops retrying and appears in plain words with Try again / Discard.
- **Built for one hand mid-set:** 56 pt touch targets, large metrics readable at arm's length, outcome buttons that open in place rather than pushing screens.
- Honest empty states ('Nothing booked', 'Nobody to chase', 'Nothing refused') and explicit online-only notice before Sell a package.

Visual system — acid lime on charcoal



bg	surface	raised	accent	success	warning	danger
#121212	#1c1c1e	#262628	#ccff00	#4ade80	#fbbf24	#fb7185

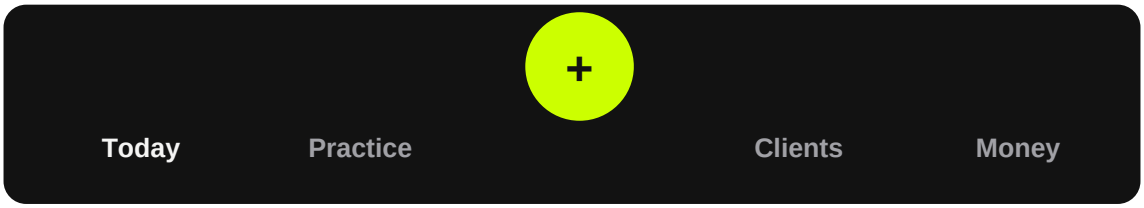
Neutrals are warm rather than blue so the yellow-green accent reads as a mark on a ground, not a competing hue. Dark text (#121212) sits on the accent, since white is illegible on it.

Token	Values
Spacing	xs 4 · sm 8 · md 12 · lg 16 · xl 24 · xxl 32
Radius	sm 10 · md 14 · lg 20 · xl 28 · pill 999 — strongly rounded cards, pill-shaped controls signal tappability
Type	Display 40/800 · Title 26/700 · Heading 17/700 · Body 16/500 · Small 14 · Caption 12 · Label 11 uppercase tracked · Metric 34/800 for loads and reps
Touch	Target 56 pt; floating action button 58 pt straddling the tab bar

Component library

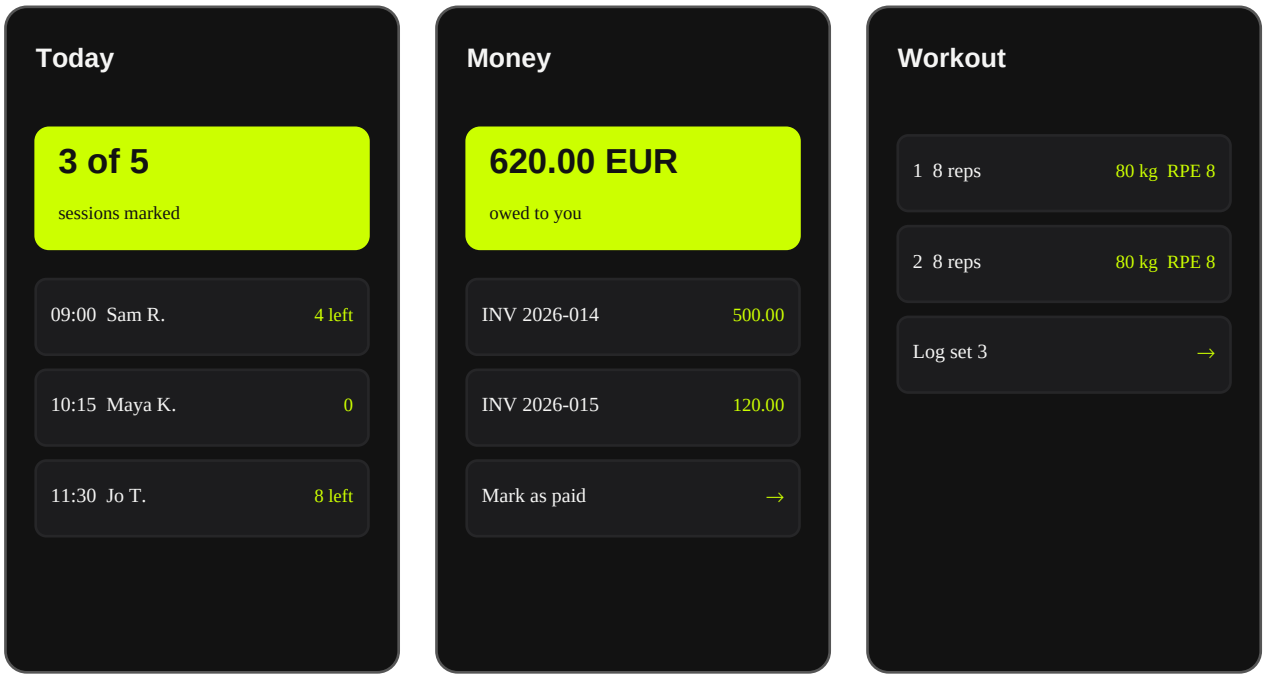
Screen, Card, Display/Title/Heading/Label/Body/Caption, Metric, Button (primary/danger/quiet), TextButton, Pill and Chip (success, warning, danger, muted), Field and NumberField, SegmentedChoice, Banner, Progress, Empty, Row, Divider, and the sync badge.

Navigation



Four tabs — **Today, Practice, Clients, Money** — with a lime floating '+' action button in the centre. Stack routes sit above: sign-in, client/[id], client/new, sell-package, workout/[id], sync, and a not-found page.

5. Screen-by-screen



Schematic wireframes illustrating layout and hierarchy, not screenshots.

Screen	Purpose & behaviour
Sign in / Create account	Email, password; sign-up adds name, business name, currency. Toggle 'Start a new practice' / 'I already have an account'.
Today	The roster: time, client, credits-remaining chip, status pill. Outcome buttons (Completed, No-show, Late cancel, Early cancel) open in place. Quick actions: Log workout, Profile. Header shows avatar and a 'sessions marked' progress figure.
Practice (dashboard)	Hero metric 'earned this month' plus today, left this week, unpaid invoices; list of clients to renew.
Clients	Searchable roster ('Name, email or phone') with credits chip; empty states for no clients / no match. Add client via client/new.
Client profile	Credits left (or 'overdrawn'), receivables, measurements with Record → weight, body fat, Save measurement, recent volume. Actions: Start a workout, Sell a pack.
Sell a package	Description, Sessions, Price each; total 'to collect'; Issue invoice, then 'Send the link' for WhatsApp. Online-only, stated up front.
Workout logger	One exercise at a time via chips and '+ Exercise' finder; Reps / kg / RPE fields; 'Log set N'; 'Repeat last time'; set rows show reps, load, RPE and done mark; Finish workout.
Money	Chase list with ageing; Mark as paid with prefilled amount, method choice, reference (bank reference, envelope, cheque number); Record payment.
Sync	What is waiting and what was refused, in plain words; Sync now, Try again, Discard, Sign out. A corner badge counts pending operations and flags 'needs attention'.

6. Architecture & quality

- **Repository layout:** cmd/ (api, worker, migrate); internal/ (platform, db, ledger, billing, crm, scheduling, programming, sync, auth, dashboard, media); app/ (Expo client); docs/adr/.

- **ADRs:** 0001 double-entry ledger · 0002 tenant isolation via RLS · 0003 encryption-at-rest scope · 0004 offline conflict policy · 0005 Expo single codebase · 0006 invoice PDF via headless Chromium.
- **API:** ~45 REST endpoints under /v1 documented in OpenAPI (auth, sync, clients, sessions, credits, invoices, receivables, dashboard, payment methods, exercises, programs, workouts) plus /healthz, /readyz and /public/invoices/{token}.
- **Testing:** unit, RLS/ledger/journey integration tests against real Postgres, property tests on ledger sequences, client typecheck + bundle of every route, and a live end-to-end test that runs the real sync engine against a server — which caught five bugs both suites had missed.
- **Run:** `make up`, `make run`, `make app-start`; demo mode needs no server.

Known gaps: booking a session has no in-app screen yet (API only).